INF2009F -- 2023 Problems in Software Development & Software Development Methodologies Exercise 1 ---- 15 Marks	
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Marks	/15

IMPORTANT NOTE:

- **Copied or Plagiarised answers will get 0 and student will be referred to Elsje Scott and Ayanda Pekane for further actions**
- **Please upload your completed exercise on Vula by the 20th February 2023 at 9:00am**

1. **Explain in your own words, why IT Projects fail (3 marks)**

IT Projects can fail for many reasons. The team involved may fail to deal with risks associated with the project, often due to the risks remaining undetected for too long or the risks may be difficult to deal with due to the complexity of IT Projects. The complexity also increases the likelihood of undetected risks. The scope of a project may not be specified clearly, meaning the requirements and deadlines are unclear. As a result, the project is underestimated and there is insufficient resources for the project, or deadlines can be missed. With unclear requirements, it can be difficult to determine whether the project has succeeded and thus can lead to projects failing. Missed deadlines delay the outcome, requiring more resources than planned for and may result in overworked employees and lower quality work. Alternatively, a project can fail if communication is poor. This results in uncertainty and important details for the project not being conveyed as necessary. Communication is also key to convey the project scope so deadlines are not missed. With the complexity of IT Projects, effective communication is necessary for success.

2. **What are the disadvantages of using a plan-based approach (Traditional SDLC) to software development? (Describe each disadvantage in your own words) (3 marks)**

A plan bases approach involves working on a project in different specified phases. The first disadvantage associated with this is as a transition between phases occurs, it is possible that such transition takes long. Another risk in this transition is that some information (from previous phases) can go missing and requires more time to deal with this problem. This can extend the duration of the problem, and possibly create redundant work when the information is lost. Thus as the project progresses, the risk associated with the project increases as there is more complexity and information that must be dealt with. A second disadvantage with the plan-based approach is that after finished phases, it is difficult to change information from previous phases and with software development, change is necessary (either due to the complexity of software or the customers wanting a change in the final product). Another disadvantage of the plan-based approach is that software is only tested right before they implement it and thus the testing is done after the code is finished, and with large amounts of code. Issues that arise will be very difficult to fix, with the project being very large and complex. Moreover, software is only implemented at the very end of the project.

3. Give a summary of the Agile manifesto. (3) (Hint – do some research on this)

The Agile Manifesto is a document that introduces 12 important principles of Agile development and 4 values that should be used in the development of software. The principles focus on prioritizing the customer and supplying working code on a frequent basis, focusing on the software over documentation and welcoming change into the project at any point. It also prioritizes simplicity in code over complexity, encourages self-organising teams, communication between software developers and the business workers and encouraging reflecting on the effectiveness of one's work. The four fundamental values involve focusing on people rather than processes and tools, software over documentation, customers over contracts, and implement changes instead of following a set plan from the beginning. These values and principles help improve the development of software and allow companies to create functioning software that the customers want with a better work environment focusing on needs by the customer.

4. Use (in your opinion) 3 of the most significant of the 12 principles of Agile development to motivate why agile software development methodologies can resolve the problems introduced by plan-based (traditional SDLC) approaches? (6 marks)

Development methodologies—Mike Chapple:
https://www.linkedin.com/learning/search?keywords=Systems%20development%20methodologies%20&u=70295562

Any arguments around the elements below can gain marks for a student

1. A very significant principle of Agile development states that change must be welcomed. The problem that can arise in a plan-based approach is that in this approach, a plan created at the beginning and no changes to the project can be made after this. With software, that is often an issue changes being needed. An example of such could be wanting more functionality from a software project. The agile development sees change as welcomed for a project and allows changes to be made whenever the customer wants. This results in a better final product with greater customer satisfaction
2. A problem involving testing can arise with the traditional SDLC as software projects often require large amounts of code. Any problem brought up with testing can be extremely difficult to fix with the complexity and size of the finished code. This will consume a large amount of time. A principle of the Agile methodology is to release software frequently. The advantage of this is then is the testing will be integrated testing. The new code will be tested alone, and then tested once integrated into the rest of the code, thus they will only have to look through the code for the implementation of the new code to fix errors when testing.
3. The Agile methodology has a principle which involves prioritising customer satisfaction by involving the customer during development. A traditional plan approach involves the customers at the beginning and only then the end of the project. This can create problems because the customer cannot change what they want whilst the product is being created and will result in the customers being less satisfied. The Agile approach offers a solution by involving the customers throughout the development. They become the priority and as such, feedback on the product is provided sooner allowing for a quicker and better final product

Parameter	Traditional	Agile
Requirements	Fixed (decided at the start)	Evolving (can Change)
Customer Involvement	Before, After	During
Testing	After Code	Integrated
Feedback & Communication	After	During
Concentration On (Development process)	Processes, Reviews, Doc	Workable Software
Focus	Plan Driven	Feature Driven